East West University
Department of CSE

Course Code: **CSE 347**
Course Title: **Information System Analysis and Design**

Project: **StudyByte – Peer-to-Peer Micro-Tutoring**

Task: **Requirement Analysis and Feasibility Study**

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GROUP NO: **04**

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1. Introduction

1.1 Project Overview

StudyByte is a web-based Peer-to-Peer Micro-Tutoring Information System developed for university students. The system enables students to offer and receive topic-based tutoring sessions through a structured digital marketplace.

The platform integrates role-based access, booking and scheduling, a digital wallet system, escrow payment mechanism, session verification, and administrative monitoring to ensure security, trust, and quality learning.

This project focuses on the **Analysis and Feasibility phases** before moving to design and implementation.

1.2 Project Objectives

- To create a secure academic tutoring marketplace.
- To enable structured booking and scheduling.
- To implement a token-based wallet with escrow protection.
- To ensure verified session completion before payment release.
- To maintain tutor quality using rating and review systems.
- To provide administrative control and dispute management.

1.3 Project Scope

In Scope

- User registration and authentication
- Role-based dashboards
- Topic marketplace and search
- Booking and scheduling
- Digital wallet and escrow system
- Session verification

- Rating and review
- Admin monitoring

Out of Scope

- Built-in video conferencing
- External banking gateway integration
- Multi-institution deployment (Phase 1 limited to one university)

2. Core Features and Project Flow

2.1 Core Features

1. User Registration & Authentication
2. Role-Based Access Control
3. Profile Management
4. Topic Marketplace
5. Search & Filtering System
6. Availability & Scheduling System
7. Booking Management System
8. Digital Wallet System
9. Escrow Payment System
10. Session Verification System
11. Rating & Review System
12. Admin Control & Dispute Management

2.2 Logical Project Flow

- **User Entry & Authentication**

  User: Student (Learner/Tutor), Admin

  Logical Steps:

  1. User registers using university email.
  2. System verifies email domain.

3. User selects role (Learner or Tutor).
4. System creates profile.
5. Dashboard loads according to role.

Output: Authorized user enters system.

- **Tutor Marketplace Setup**

User: Tutor

Logical Steps:

1. Tutor logs in.
2. Tutor creates micro-topic listing.
3. Tutor sets:
    - Token price
    - Available time slots
4. System stores topic in marketplace database.
5. Topic becomes searchable.

Output: Marketplace updated with tutoring service.

- **Learner Search & Booking**

User**:** Learner

Logical Steps:

1. Learner searches topic.
2. System filters by:
    - Price
    - Rating
    - Urgency
3. Learner selects tutor.
4. Learner selects available time slot.
5. System checks wallet balance.

6. Booking request confirmed.

Output: Booking created.

- **Escrow Payment Protection**

Logical Steps:

1. Tokens deducted from learner wallet.
2. Tokens moved to escrow account.
3. Booking status updated to "Confirmed".

Important:

Tutor does NOT receive payment yet.

Output: Secure transaction locked.

- **Session Execution & Verification**

Logical Steps:

1. Session takes place.
2. System generates unique meeting code.
3. Learner provides code to tutor.
4. Tutor submits code.
5. System verifies code.

6. If valid:
    - Escrow releases tokens to tutor wallet.
    - Booking marked as "Completed".

Output: Payment transferred securely.

- **Rating & Admin Oversight**

Logical Steps:

1. Learner submits rating & review.
2. System updates tutor average rating.
3. Admin monitors:
    - Transactions
    - Disputes
    - Withdrawal requests
4. Admin can suspend accounts if necessary.

Output: Trust & governance maintained.

3. Requirement Determination

Functional Requirements

1. User Management

1.1 The system must allow students to register using university email.
1.2 The system must verify the email domain before account activation.
1.3 The system must allow secure login and logout.
1.4 The system must support three roles: Learner, Tutor, and Admin.
1.5 The system must provide separate dashboards for each role.

2. Profile Management

2.1 The system must allow users to update profile information.
2.2 The system must allow tutors to add expertise topics.
2.3 The system must store department and student ID information.

3. Marketplace Management

3.1 The system must allow tutors to list micro-topics.
3.2 The system must allow tutors to set a token price per session.
3.3 The system must organize topics by department/category.
3.4 The system must allow learners to search topics.
3.5 The system must allow filtering by price, rating, and urgency.

4. Booking & Scheduling

4.1 The system must allow tutors to set available time slots.
4.2 The system must prevent double booking of time slots.
4.3 The system must allow learners to book available sessions.
4.4 The system must store booking history.

5. Payment & Wallet System

5.1 The system must maintain a digital wallet for each user.
5.2 The system must deduct tokens upon booking.
5.3 The system must hold tokens in escrow until session verification.
5.4 The system must release tokens after meeting code validation.
5.5 The system must maintain transaction history.

6. Session Verification

6.1 The system must generate a unique meeting code per session.
6.2 The system must verify meeting code before releasing payment.

7. Rating & Review

7.1 The system must allow learners to rate tutors.
7.2 The system must calculate the average rating automatically.
7.3 The system must display rating in tutor profile.

8. Admin Control

8.1 The system must allow admin to monitor users.
8.2 The system must allow the admin to manage disputes.
8.3 The system must allow admin to approve token withdrawals.
8.4 The system must allow the admin to suspend suspicious accounts.

Non-Functional Requirements

1. Performance Requirements

1.1 The system must load pages within 2 seconds.
1.2 The system must support at least 500 concurrent users.
1.3 Wallet transactions must be processed within 3 seconds.

2. Security Requirements

2.1 Only registered users can access the system.
2.2 Only learners can book sessions.
2.3 Only tutors can list topics.
2.4 Only the admin can approve withdrawals.
2.5 Passwords must be encrypted.

3. Operational Requirements

3.1 The system must run on modern web browsers.
3.2 The system must support mobile responsiveness.
3.3 The system must maintain daily database backup.

4. Reliability Requirements

4.1 The system must ensure 99% uptime.
4.2 The escrow system must prevent payment loss.

5. Usability Requirements

5.1 The interface must be simple and user-friendly.
5.2 The booking process must not exceed five steps.
5.3 Error messages must be clear and understandable.

4. Feasibility Study

4.1 Technical Feasibility

Factor	Evaluation
Development Tools	HTML, CSS, JavaScript, Python/PHP, MySQL

Hardware Requirements	Standard web server sufficient
Team Skills	Programming & database knowledge available
Technology Availability	Required technologies readily available
System Complexity	Moderate and manageable

Conclusion: Technically feasible.

4.2 Economic Feasibility

Cost Element	Assessment
Development Cost	Low (student project)
Hosting Cost	Minimal
Maintenance Cost	Low
Expected Benefits	Academic improvement & peer earning
Cost-Benefit Ratio	Positive

Conclusion: Economically feasible.

4.3 Operational Feasibility

Factor	Evaluation
User Acceptance	High among students

Ease of Use	Simple interface
Integration with Campus	Suitable
Administrative Support	Manageable

Conclusion: Operationally feasible.

4.4 Schedule Feasibility

Week	Activity
Week 1	Requirement Analysis
Week 2	System Design
Week 3	Database Design
Week 4	User Module
Week 5	Booking Module
Week 6	Wallet & Escrow
Week 7	Rating & Admin
Week 8	Testing & Documentation

Conclusion: Achievable within semester timeline.

4.5 Legal Feasibility

- No violation of data protection regulations.
- Tokens represent internal virtual currency.
- The system complies with university academic policies.
- Admin ensures ethical usage.

Conclusion: Legally feasible.

Final Conclusion

The StudyByte system has clearly defined functional and non-functional requirements and has been evaluated under all feasibility categories.

The project is technically, economically, operationally, legally, and schedule-wise feasible and suitable for development as a CSE347 Information System project.